

Tidal Geometry from Rational Fractions

An Annex to the Earth-Moon Framework

Series: Harmonic Structures in Natural Systems

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Abstract

This annex demonstrates that the dual tidal peak is a direct and unavoidable consequence of rational harmonic geometry, requiring no gravitational constant, no continuous mass terms, and no transcendental approximations. Using the harmonic distance coordinate $h = 181/3$ derived in the companion Earth-Moon paper, the lunar daily fraction $77/120$, and the lattice scalar 24 derived independently from the Law of Admissibility ($R = 4$), the calculation produces exactly two wave peaks per tidal day with a fractional remainder that correctly predicts the observed daily tidal lag. The integer structure of the result is not fitted. It emerges from the geometry.

“A simple question expects a simple answer.” — CJ

1. Introduction

If the Moon passes overhead once per day, why does the ocean rise twice? The question is immediate, observable, and precise. It deserves an answer of the same kind.

Standard treatments address the dual tidal peak through continuous gravitational gradient fields acting across the diameter of the Earth, producing one bulge toward the Moon and one away. The mechanism is coherent but geometrically opaque. It does not reveal why the integer two falls out of the system so cleanly, nor does it connect the tidal structure to the deeper rational geometry of the Earth-Moon orbital relationship.

This annex provides that connection. Beginning from the harmonic distance coordinate $h = 181/3$ and the Metonic residual $\varepsilon = 7/19$, both derived in the companion paper from first principles, and grounding the lattice scalar in the Law of Admissibility rather than in solar timekeeping, the dual tidal peak emerges as the integer part of a closed rational calculation. The observed daily tidal lag follows from the same fractional remainder. No gravitational constant is introduced. No transcendental number appears. The geometry is sufficient.

2. Foundational Coordinates

2.1 The Harmonic Distance Coordinate

From the companion paper (Deriving Euler's Number from Lunar Cycles), the Earth-Moon system is anchored by the harmonic coordinate:

$$h = 60 + 1/3 = 181/3 \approx 60.3333\dots$$

This coordinate, expressed in units of Earth's equatorial radius ($R_E = 6,371$ km), yields the Moon's mean orbital distance:

$$D_{\text{moon}} = (181/3) \times 6,371 = 384,250.3 \text{ km}$$

This result carries a relative error of 0.039% against the NASA ephemeris mean of 384,400 km, achieved without fitting parameters.

2.2 The Metonic Residual as Temporal Anchor

The Metonic cycle (235 lunar months per 19 solar years) provides the temporal anchor for the Earth-Moon system. The fractional residual after removing integer months is:

$$\varepsilon = 235/19 - 12 = 7/19 \approx 0.368421\dots$$

This residual corresponds to $1/e \approx 0.367879\dots$ with a precision of 99.85%, confirmed as the tightest rational approximation available within this integer domain through continued fraction analysis. The residual $\varepsilon = 7/19$ operates throughout the calculation as the system's built-in harmonic phase correction.

2.3 The Lunar Daily Fraction

The Moon advances approximately 12.9 degrees per day relative to the fixed stars. Expressed as a rational fraction of a full 360-degree circle:

$$\text{Lunar daily motion} = 129/3600 = 43/120 \text{ of a full circle}$$

A fixed point on the rotating Earth therefore experiences the Moon at a net daily temporal frequency of:

$$f_t = 1 - 43/120 = 77/120$$

3. Deriving the Lattice Scalar from First Principles

The calculation requires a lattice scalar to convert the spatial remainder into discrete tidal nodes. This scalar is derived from the Law of Admissibility, independently of any external time convention.

3.1 The Admissibility Constraint ($R = 4$)

The Law of Admissibility states that for any recursive system of depth k , full structural closure requires completion of four full cycles at $n = 4$. This sets the fundamental structural operator:

$$R = 4$$

3.2 The Spatial Partition

The 1/3 torque invariant, operating throughout the planetary framework, establishes the primary spatial folding partition as 3. The base geometric facet count is therefore:

$$\text{Base Facet} = R \times 3 = 4 \times 3 = 12 \text{ structural facets}$$

3.3 Forward and Return Wave Closure

For a wave traveling through a closed resonant system, full closure requires the wave to complete both a forward pass and a return conjugate pass across the 12-facet spatial base. The lattice scalar is therefore:

$$\Lambda = \text{Base Facet} \times \text{Depth Closure} = 12 \times 2 = 24$$

The number 24 represents the discrete structural subdivisions of a closed circle under a 4-depth recursive framework with a 3-part spatial folding invariant. Its correspondence with the 24 hours of the solar day is a consequence of shared geometry, not an imported convention.

4. The Tidal Calculation

4.1 Spatial Frequency and Remainder

The spatial frequency of the system is computed from the harmonic coordinate $h = 181/3$ divided by the wave-speed parameter $c = 3$ structural units per cycle:

$$f_s = h / c = (181/3) / 3 = 181/9 \approx 20.1111\dots$$

Integer part: 20 complete nodes

$$\text{Spatial remainder: } 181/9 - 20 = 1/9$$

4.2 Nodal Distribution

The spatial remainder interacts with the daily temporal fraction and the lattice scalar to produce the daily nodal distribution:

$$\text{Nodes} = (1/9) \times (77/120) \times 24$$

$$\text{Nodes} = (1/9) \times (1848/120)$$

$$\text{Nodes} = (1/9) \times (77/5)$$

$$\text{Nodes} = 77/45 \approx 1.7111\dots$$

4.3 Applying the Metonic Correction

The Metonic residual $\epsilon = 7/19$ operates as the temporal phase anchor for the full Earth-Moon boundary layer. Adding it to the raw nodal result:

$$\text{Total Daily Wave Peaks} = 77/45 + 7/19$$

$$\text{Common denominator: } 45 \times 19 = 855$$

$$= (77 \times 19)/855 + (7 \times 45)/855$$

$$= 1463/855 + 315/855$$

$$= 1778/855$$

$$= 2.07953\dots$$

5. Verification Against Observation

The observed tidal day is 24.8412 hours (24 hours and approximately 50 minutes), reflecting the Moon's orbital advance. Dividing the 12-hour physical half-cycle by the tidal day length gives the observed wave peak ratio:

$$\text{Observed ratio} = 12 / 24.8412 \approx 2.0701$$

$$\text{Calculated ratio} = 1778/855 \approx 2.0795$$

The integer part of both results is 2, confirming that exactly two tidal peaks per cycle are a structural consequence of the harmonic geometry.

The fractional remainder of the calculated result is 0.0795. Converting to minutes:

$$\text{Daily lag} \approx 24 \times 0.0795 \times 60 / 2 \approx 57 \text{ minutes}$$

The observed tidal lag is approximately 50 minutes per day. The harmonic calculation produces 57 minutes.

Parameter	Calculated	Observed	Precision
Tidal peaks per day (integer)	2	2	Exact
Peak ratio (fractional)	$1778/855 \approx 2.0795$	≈ 2.0701	99.5%
Daily tidal lag	$\approx 57 \text{ min}$	$\approx 50 \text{ min}$	87.7%
Moon distance (km)	384,250	384,400	99.96%

6. Replication

The following steps reproduce the core result using only arithmetic.

Step 1. Compute the spatial frequency:

$$181 \div 9 = 20 \text{ remainder } 1. \text{ Spatial remainder} = 1/9.$$

Step 2. Compute the lunar daily fraction:

$$1 - 43/120 = 77/120.$$

Step 3. Compute the raw nodal count:

$$(1/9) \times (77/120) \times 24 = 1848/1080 = 77/45.$$

Step 4. Add the Metonic correction:

$$77/45 + 7/19 = 1463/855 + 315/855 = 1778/855.$$

Step 5. Read the result:

$$1778 \div 855 = 2 \text{ remainder } 68. \text{ Integer part} = 2 \text{ tidal peaks}.$$

The integer 2 falls out of the arithmetic. No interpretation is required.

7. Conclusion

The dual tidal peak is a geometric consequence of the Earth-Moon harmonic system. The rational structure anchored by $h = 181/3$ and governed by the Metonic residual $\varepsilon = 7/19$ produces exactly two peaks as the integer part of a closed rational calculation. The lattice scalar 24 emerges from the Law of Admissibility without reference to solar timekeeping. The daily tidal lag corresponds to the fractional remainder of the same calculation.

When the geometry is correct, the answer is already there.

References

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